PROJECT:

1X45MW Captive Power Plant, (Phase-I) at Jamshedpur, Jharkhand

DOCUMENT TITLE:

BATTERY CHARGER & BATTERY SIZING CALCULATION

CLIENT	M/S. Gajanan Ferro Pvt Ltd.,
PROJECT CONSULTANT	M/s. Aquatherm Engineering Consultant India Pvt Ltd.,
PROJECT NUMBER	11894
DOCUMENT NUMBER	400-ENG-11894-035

PROJECT CONSULTANT:

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Project	**1X45MW Captive Power Plant, (Phase-1) at Jamshedpur, Jharkhand,**		
Client	**M/s. Gajanan Ferro Pvt Ltd.,**		
Consultant	**M/s. Aquatherm Engineering Consultants (India) Pvt. Limited**		
Document	**Battery Charger & Battery Sizing Calculation**		
Document Number	**400-ENG-11894-035**	**Rev**	**P**

Revision Summary

Rev. No	Date	Prepared By	Checked By	Approved By	Revision Details
P	01.07.2025	V.R.I	S.P	R.A	First Issue

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I. DC Load List - Phase - 1:-

S.No	Panel / Load Description	Qty Per Panel	Total Qty	(A) Continuous Each Load in Watts	Continuous Total Load in Watts	(B) Emergency Each Load in Watts for 1 Hour	Emergency Total Load in Watts	(C) Intermittent Each Load in Watts	Intermittent Total Load in Watts	Remarks
A	**STG Panel**									
1	45MW - STG DC Loads	1	1	2273	2273	16667	16667	33284	33284	
	Sub Total				**2273**		**16667**		**33284**	
B	**33kV Switchgear Panel**	**15**								
1	Indicating Lamps	5	75	2	150	0	0	0	0	
2	DC Supervision Relay	2	4	6	24	0	0	0	0	
3	Feeder Protection Relay	1	15	18	270	0	0	0	0	
4	Trip Circuit Supervision Relay	1	15	5	75	0	0	0	0	
5	Master Trip Relay	1	15	0	0	0	0	150	2250	
6	Trip Coil	2	30	0	0	0	0	250	7500	
7	Closing Coil	1	15	0	0	0	0	250	3750	
8	Annunciator	24	24	0	0	0	0	0.5	12	
9	Hooter	1	15	0	0	0	0	2	30	
10	Digital Ammeters	1	4	3	12	0	0	0	0	
11	Digital Voltmeters	1	3	3	9	0	0	0	0	
12	MFM	1	13	5	65	0	0	0	0	
13	Auxiliary Relays / Contactors	5	75	6	450	0	0	0	0	
	Sub Total				**1055**		**0**		**13542**	
C	**415V PCC**	**16**								
1	Indicating Lamps	5	80	2	160	0	0	0	0	
2	DC Supervision Relay	1	2	6	12	0	0	0	0	
3	Master Trip Relay	1	3	0	0	0	0	150	450	
4	Closing Coil	1	2	0	0	0	0	250	500	
	Sub Total				**172**		**0**		**1450**	
D	**Emergency Lighting Load**									
1	Lamp	25	25	0	0	100	2500	0	0	
	Sub Total				**0**		**2500**		**0**	
F	**Control & Relay Panel**									
1	DC Loads	2	2	500	1000	0	0	100	200	
	Sub Total				**1000**		**0**		**200**	
	Grand Total in Watts				**4500**		**19167**		**48476**	
	Total Load in Amps				**21**		**88**		**221**	

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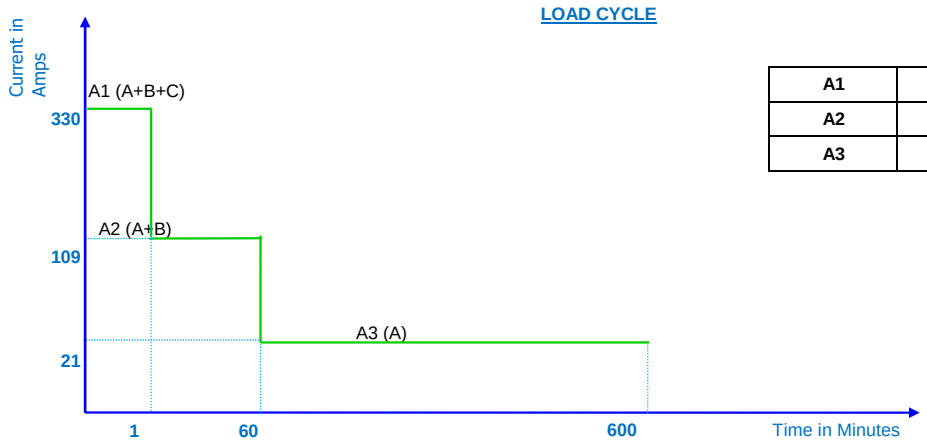
II. Battery Sizing Calculation - Unit-1

LOAD CYCLE



A1	**=**	**A+B+C**	**=**	**330**
A2	**=**	**A+B**	**=**	**109**
A3	**=**	**A**	**=**	**21**

Input Parameters

Type of Battery	:	VRLA	
System Nominal Voltage	:	220	Volts
Cell Voltage	:	2	Volts
Minimum Operating Voltage	:	198	Volts
Maximum Operating Voltage	:	242	Volts
Operating Temperature	:	27	°C
Design Margin	:	1.15	
Ageing Factor	:	1.25	
Temperature Correction Factor	:	1	
End Cell voltage	:	1.85	Volts / Cell
Number of Cells	:	110	

IEEE Std 485 Method of Battery Sizing

(1)	(2)	(3)	(4)	(5)	(6)	(7)
Period (Nos)	**Load (Amps)**	**Change in Load (Amps)**	**Duration of Period (Mins)**	**Time to End Section (T) (Mins)**	**Capacity at T min Rate K factor**	**Req. Size [(3)x(6)] (AH)**
Section 1 - First 1 Period only - If A2 > A1, then go to section 2						
1	A1 = 330	A1 - 0 = 330	M1 = 1	T = M1 = 1	0.5	165
					Section 1 Total	**165**
Section 2 - First 2 Periods only - If A3 > A1, then go to section 3						
1	A1 = 330	A1 - 0 = 330	M1 = 1	T = M1 +M2= 60	1.88	620.4
2	A2 = 109	A2 - A1 = -221	M2 = 59	T = M2= 59	1.88	-415.48
					Section 2 Total	**204.92**
Section 3 - First 3 Periods only - If A3 > A2, then go to section 4						
1	A1 = 330	A1 - 0 = 330	M1 = 1	T = M1 +M2+M3 = 600	11.04	3643.2
2	A2 = 109	A2 - A1 = -221	M2 = 59	T = M2+M3= 599	11.04	-2439.84
3	A3 = 21	A3 - A2 = -88	M3 = 540	T = M3= 540	10.04	-883.52
					Section 3 Total	**319.84**

Required Capacity (Maximum Among Sections 1, 2 & 3)	**320**	AH
After Ageing Factor Correction (Required Capacity x Ageing factor)	**400**	AH
After Design margin considered (After Ageing Correction x Design Margin)	**460**	AH
Minimum Required Capacity	**460**	AH to 1.85 ECV
Recommended Battery Capacity	**500**	**AH**

III. Charger Sizing Calculation - Unit-1

Float Charger

Float Charger Current Rating — Continuous Load Current + Float Charging Current of Battery(1mA/AH)

21 + (400 x 1/1000)

21.5 Amps

Consider 20% Design margin — 27 Amps

Boost Charger

Case - 1

Boost Charger Rating — Min 25% of Battery AH capacity

500 x 0.25

125 A

Case-2

Boost Charger Rating — Load Current (Continuous+Emergency) + Boost Charging Current of Battery (10%)

159 Amps

159 Amps

Recommended FCBC Current Rating	**200**	**A**

Note:-

1. DC EOP motor rating is preliminary. If any change on the same will impact the battery and battery charger capacity. The rating of battery and charger will be firmed up after the receipt of the same from STG vendor.
2. The selected battery charger shall be suitable for 9.3kW EOP DC Motor & 3.7kW JOP DC Motor starting and running.
3. T_{min} rate "K" factors of batteries are taken from reputed manufactures catalogue.